Field-Specific Evidence Contracts for Robot Feedback Monitoring: A Retrospective Multi-Schema Study

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Abstract

Robot feedback monitors check message timing, numerical availability and kinematic behavior, but a single verdict can conceal which diagnostic premise is missing. We formulate a reusable, field-specific evidence contract that declares required joints, fields, producer flags, native units and clock continuity. It returns property-specific results with explicit abstention. We evaluate it retrospectively through paired predicate comparisons on unchanged operational records, without physical fault labels. In six incident arm traces, all 1,968 rows pass a 20 ms recorded-age cutoff; two traces exceed their configured speed bound, while a different trace holds an identical position–velocity vector for 0.664 s below that bound. At a common 0.5 s horizon, its integral discrepancy is 0.935 degrees, versus at most 0.00934 degrees in 311 direct contextual windows. Another 38 position-eligible windows lack required velocity and remain undetermined. Among 77,627 archived joint-state messages, field-specific selection retains finite arm inputs in 54,897 messages rejected by a message-wide check. Conversely, producer flags withhold six numerically finite, zero-velocity rows in three bench records. One shared validator handles four schema families and passes 27 software invariant checks. The contribution is an executable comparison of evidence requirements and record-level disagreements, rather than a new temporal logic or calibrated fault detector. The persistent inconsistency remains one selected case; physical acquisition freshness, root cause and cross-platform diagnostic accuracy remain unverified.

Keywords: Robot diagnostics, Data provenance, Runtime monitoring, Joint feedback, Timestamp freshness, Fault observability

Article type: Original research. **Categories:** (1), (3), (6). **MSC 2020:** 68T40, 93C85.

1 Introduction

Robot software connects sensing, learned policies, trajectory controllers and hardware interfaces through several asynchronous layers. The Robot Operating System 2 (ROS 2) makes these layers easier to assemble and deploy, but the resulting execution record reflects both physical processes and software transformations [1]. An execution monitor therefore faces two different questions: whether a message has recently arrived, and whether its contents represent a recently acquired and usable measurement. These questions become difficult to separate when a controller returns success, the next action is rejected as unsafe, and joint feedback continues to carry advancing timestamps.

The distinction matters for diagnosis as well as control. A constant position may describe an arm holding its pose, a quantized measurement or an unchanged cached value. A nonzero velocity paired with constant position indicates a stronger inconsistency, but does not identify which field or layer is wrong. An accepted action is evidence about a software transaction; it is not an independent measurement of motion. Treating these observations as interchangeable can turn a useful protection message into an unsupported physical or causal explanation.

Prior empirical work already establishes the value of combining observability data. Song et al.

investigated robotic bugs using logs, traces and trajectories [2]. ROBUST provides reproducible cases for studying faults in ROS software [3]. Runtime verification and tracing provide mechanisms for checking events and reconstructing execution [4, 5]. These contributions motivate a question about shared measurement provenance. What can a combined record establish when several monitors consume the same uncertain state? Additional views need not provide independent physical observations.

We address the gap with a field-specific evidence contract: a small, executable specification of the inputs and assumptions needed for each recorded property. It separates numerical eligibility, producer-declared availability, clock continuity and internal kinematic evidence. Missing prerequisites yield an undetermined result for that property, while other supported properties remain available. The implementation is independent of robot names and is reused through declarative adapters for four schema families.

Three questions organize the study. **RQ1:** Which disagreements appear when recency, speed-bound, position-only and field-validity predicates examine the same retained records? **RQ2:** Can a shared contract preserve explicit eligibility and abstention across flat CSV, grouped telemetry, named JointState arrays and producer-flag bench logs? **RQ3:** Which physical conclusions remain unsupported after those checks and control events are combined?

The contributions are (i) a reusable contract and reason-coded implementation for property-specific record validation; (ii) paired comparisons of explicit monitoring predicates, including equal-duration kinematic windows and both permissive and overbroad validity decisions; and (iii) a reproducible schema-portability evaluation with software invariant checks. The material includes an incident archive, historical traces from a second robot-day, metadata-selected JointState archives and three distinct bench records. It supports empirical comparisons of recorded evidence, without converting stored copies or message counts into independent physical trials. The implementation, adapters, inputs and complete comparison outputs are supplied for offline reuse.

2 Related work

2.1 Robotic fault diagnosis and empirical evidence

Fault diagnosis can use model residuals, signal processing or learned representations. Adaika et al. review these approaches for unmanned aerial vehicles, including their deployment constraints [6]. Cacace et al. combine external-wrench estimation and a recurrent network for motor fault detection and isolation, with training and testing in simulation [7]. Such studies evaluate a decision rule against specified fault conditions. Here, independently verified physical labels are unavailable. Reporting detector accuracy would therefore conflate agreement with an incident narrative and verification of the underlying fault.

The closest methodological precedent is observability-based empirical diagnosis [2]. We do not claim that correlating logs, trajectories and events is new. Our narrower focus is the provenance of the observations being correlated. Fields in one message can have different update histories, and internal kinematic consistency does not supply independent physical truth. ROBUST emphasizes inspectable, reproducible bug artifacts [3]; our archive instead represents an operational incident with incomplete acquisition and command coverage. Keeping that difference explicit prevents a retrospective record from being presented as a controlled fault benchmark.

2.2 Tracing and runtime monitoring

The ros2_tracing framework exposes computation and communication behavior that ordinary message recording cannot fully reveal [4]. Caldas et al. develop guidelines for runtime verification and field testing, including timing, isolation and oracle definition [5]. Ogma translates structured requirements into monitors integrated with ROS 2 [8], while ROSMonitoring 2.0 extends monitoring across services and ordered topics [9]. RTAMT evaluates temporal specifications quantitatively and integrates with ROS [10]. These methods make the monitored property explicit. Our analysis concerns the measurement semantics needed to evaluate a property from joint telemetry, rather than learning task-level success labels.

Our use of an indeterminate outcome follows the established principle that partial observations need not justify a binary verdict. Three-valued runtime semantics formalize this distinction for temporal properties [11]. We use it at the level of diagnostic claims, without proposing a new temporal logic: a software event may be established while a physical explanation remains undetermined.

2.3 Practical monitoring predicates and the remaining gap

The ROS diagnostic updater provides frequency and timestamp checks; its timestamp implementation compares the current clock with a supplied stamp [12]. These checks answer communication-timing questions. They do not inspect whether a particular joint field changed or whether a supplied timestamp identifies its device acquisition. Configured speed bounds answer another question: whether reported velocity exceeds a limit. A position-only small-span rule describes holding, but cannot by itself distinguish holding from unavailable or inconsistent feedback.

Joint selection is already supported by the joint-state broadcaster, including selection of joints and interfaces and mapping of derived interfaces into standard fields [13]. We therefore do not claim that selecting joints, checking finite values, or returning a third truth value is novel. The gap examined here is their composition into an inspectable contract whose prerequisites, withheld calculations and disagreements with simpler predicates are evaluated on the same operational records. The comparisons below reimplement explicit predicates; they are not head-to-head reproductions of complete ROS packages, temporal-logic engines or learned fault detectors.

2.4 Timeliness, measurement provenance and action semantics

Age of information describes timeliness using the generation time of a status update [14]. Applying this concept to a robot requires identifying what was generated: an underlying device measurement, a hardware-interface snapshot, or a middleware message. Our recorded age subtracts a hardware abstraction layer (HAL) timestamp from the receiver clock. We call this quantity *recorded age*, because its numerical value does not establish the acquisition time of every field.

Controller interfaces also expose specific semantics. The joint trajectory controller documents action success relative to configured tolerances and continued use of the last commanded point [15]. Consequently, success alone does not establish subsequent standstill under another monitor's threshold. More generally, dependable-computing terminology distinguishes faults, erroneous internal states and service failures [16]. We preserve this separation: a repeated vector and a rejected policy step are observable; their unique causal explanation is not.

3 Materials and methods

3.1 Study design, archive and selection

The material was collected during an existing investigation on 26 September 2026. The archive reanalysis was performed on 19–20 October 2026. Sessions are pseudonymized as R1–R6 in chronological order. All involve one dual-arm robot with seven joints per arm and a battery-picking policy prompt. The recorded action chunk contains 20 samples; the model identifier and training-step metadata are shared across sessions. The archive contains no checkpoint or training data for independent policy verification. The configured nominal planning velocity was $2^\circ s^{-1}$, the feedback hard limit was $4^\circ s^{-1}$, and the pre-step stationary threshold was $0.25^\circ s^{-1}$. Contact-control mode was recorded as disabled. These are configuration facts, not measured performance guarantees.

Environment snapshots record software development kit (SDK) versions; detailed feedback is stored in comma-separated-value (CSV) files. Table 1 summarizes all six copied sessions. The historical directory inventory lists 11 run directories; only six have the copied session artifacts used here. The other five were excluded rather than reconstructed from narrative logs. This is convenience selection by the earlier investigation, not consecutive enrollment or random sampling. The dataset is not an estimate of incident prevalence.

Each copied session contains a run configuration, environment snapshot, step records, summary, control events and driver journal. Across these files, 25 steps were completed, four failed

Table 1 Copied session inventory. Completed steps denote software completion, not independently verified manipulation success. Raw telemetry covers only the named selected step; the remaining steps are represented by stored summaries and events

Session	SDK	Configuration	Session status	Steps	Completed	Detailed bilateral CSVs
R1	0.52.1	A	Failed	1	0	Not copied
R2	0.52.1	A	Failed	5	4	Step 5
R3	0.52.1	A	Failed	4	3	Step 4
R4	0.52.1	A	Failed	1	0	Not copied
R5	0.52.1	A	Stopped	16	16	Not copied
R6	0.56.4	B	Failed	3	2	Step 3
Total				30	25	3 paired windows, 6 CSVs

during command execution and one was rejected during the safety phase. The 119 control events are heterogeneous event records, not 119 independent trials. Detailed feedback comprises 1,968 rows from R2 step 5, R3 step 4 and R6 step 3. The R2 CSVs were preserved within a replay preparation directory. Their original run metadata and a historical remote SHA-256 for the right-arm CSV establish that they are copies of the earlier execution, not a new replay trial. Five archived remote-file checksums, covering three Python sources and two right-arm CSVs, were recomputed and matched. The other copied files have newly recorded hashes; these document present artifact identity rather than retrospective remote attestation.

R1–R5 share SDK version 0.52.1 and configuration digest A; R6 uses version 0.56.4 and digest B. The digest change, sequential operation and unrecorded physical conditions preclude attributing differences to the SDK update. Source snapshots explain parsing and exception paths. They do not establish exact correspondence between every source line and each installed binary.

3.2 Additional operational context and read-only retrieval

Two historical teleoperation capture archives, S1 and S2, were identified on a second seven-joint-per-arm robot. Both were recorded on 13 October 2026 with SDK 0.61.3; S2 followed a reboot. They constitute one additional robot-day, not two independent replications. Capture manifests, event records and collector source document ROS best-effort subscriptions and direct message serialization. They do not guarantee complete delivery or certify physical health. These two archives were retrieved through read-only file transfer. A subsequent candidate-archive census used read-only file access and low-priority directory enumeration; it introduced no subscriptions, collection programs or robot-state changes.

After reading metadata and small file headers, but before inspecting body samples, we fixed 16 nonoverlapping 4 MiB blocks at uniformly spaced byte offsets in each archive. The two source files total 3,512,779,947 bytes; 134,217,728 bytes were sampled. Source size and modification time were unchanged around every block read, and local block hashes were recorded. These checks document the retrieved segments; they are not whole-file remote checksums. Incomplete boundary lines were discarded. No analysis bridges blocks, and byte-stratified sampling is not uniform sampling in time. One S1 block contained no eligible arm-state message and remains in the coverage record.

We extract canonical seven-joint arm vectors separately from each arm's direct group-state topic and the aggregated `/joint_states` topic. An aggregated message contributes one view per arm; these views are dependent on the direct messages and are never pooled as independent measurements. Here, direct denotes the group-state topic rather than a certified device acquisition. Topic-graph snapshots identify a driver publisher, but per-message publisher identifiers are empty in the capture; continuous source attribution is unavailable. All position and velocity arrays are checked for required names, length and finite entries. An empty velocity array remains missing. It is not replaced with zero or classified as stationary.

For contextual consistency comparisons, reception-time bins are 0.5 s long, anchored at

the first arm sample of each block/topic. A position window requires at least 0.4 s of observed span, finite positions, strictly increasing reception time and no gap above 0.1 s. An integral residual additionally requires finite velocity in every row. Missing-velocity windows remain in the position-window denominator and receive an undetermined consistency outcome. Gap limits of 0.05 and 0.5 s are sensitivity settings. Residence runs also break at absent fields, nonpositive time differences or gaps above 0.1 s. These are descriptive eligibility rules rather than calibrated fault thresholds.

3.3 Archived-message audit and joint-specific eligibility

An additional audit used existing MCAP archives containing standard `sensor_msgs/msg/JointState` messages. Within the initially inspected readable pool, we selected the first lexical path per storage controller, recorded Coordinated Universal Time (UTC) day and schema digest. Selection used no kinematic outcome. Archive locations do not establish independent physical acquisition systems; source mappings and the complete candidate order remain in the evidence ledger.

Archives up to 32 MiB were copied completely. Larger archives used three five-second intervals centered at 10%, 50% and 90% of recorded duration, merging overlaps. Source size and modification time were checked around extraction. Empty intervals were retained. Exact byte ranges, hashes, extraction limits and the complete selection ledger are documented in Online Resource 1. Portable inputs preserve message order, separate clock differences and nonfinite-value masks.

The JointState interface permits optional arrays to be empty [17]. Numerical completeness must therefore be evaluated for the fields required by a specified property. For a joint set $\mathcal{J}_a$, define

$$C_a(i) = \prod_{j \in \mathcal{J}_a} \mathbb{1}\{q_{ij} \text{ and } v_{ij} \text{ are finite}\}. \qquad (1)$$

The predicate is zero if required names are absent, names are duplicated or position/velocity lengths differ from the name-array length. Names are matched through two explicit conventions, including interleaved left/right layouts, without inferring physical axes from array order. We compare all named joints (C_m) with seven names per arm ($C_L C_R$). This establishes structural eligibility; units, producer validity and acquisition provenance require separate checks.

MCAP distinguishes recording and publication times, permits publication time to default to recording time, and defines sequence as optional [18]. These fields are inspected separately from message-header time and are not assumed to identify device acquisitions. Residence calculations use the context audit's positive-gap rules and retain the complete parameter grid. They describe repetition without applying the incident angular-speed alarm threshold. The completeness comparison was added after initial inspection revealed nonfinite entries and remains a descriptive secondary analysis.

A supporting audit uses three existing bench logs to check producer-declared velocity availability, mixed angular/linear units and repeated controller-cycle identifiers. Decompressed hashes identify cross-storage copies, which are counted once (Online Resource 1).

3.4 Signals and timestamp semantics

Each incident telemetry row records seven positions, velocities, efforts, currents, temperatures and external efforts: 42 signal fields per arm. Our primary kinematic payload is the 14-component vector

$$\mathbf{x}_i = [\mathbf{q}_i^\top, \mathbf{v}_i^\top]^\top, \quad \mathbf{q}_i, \mathbf{v}_i \in \mathbb{R}^7. \qquad (2)$$

For the incident and S1/S2 arm vectors, positions and velocities are stored in radians and radians per second. Display values use degrees and degrees per second. Numerical identity means equality of the parsed decimal values at the archive's stored precision; it is not a claim of bitwise identity of the original transport messages.

The source snapshot documents `source_mono_ns` as the HAL monotonic timestamp passed through the message, `header_stamp_ns` as the message stamp, and

`recv_mono_ns` as local reception time. Define

$$a_i = (r_i - s_i)10^{-9}, \qquad \delta_i = (r_i - r_{i-1})10^{-9}, \quad (3)$$

where r_i and s_i are receiver and HAL timestamps in nanoseconds. For every incident CSV row, the saved age equaled $r_i - s_i$. We checked positive, strictly increasing timestamps within each window and did not subtract monotonic clocks across sessions or reboots. Header and wall-clock fields were used only for local consistency and event ordering. No claim of independently calibrated device-to-host clock synchronization is made.

The receiver code checks how recently a state message was received. A separate source-time field is exposed, but no original device acquisition counter, per-field acquisition timestamp or validity marker is present in the analyzed CSVs. We therefore do not rename the HAL timestamp as an independently verified sensor sampling time. Anonymous derived CSVs preserve integer timestamp differences and signal values, with absolute clock origins removed. In S1/S2, Data Distribution Service (DDS) source time, header time, HAL time and reception time remain distinct; the aggregated topic has no HAL timestamp. An absent timestamp is retained as missing, rather than being compared across clock domains.

3.5 Reusable contract and paired comparison protocol

A contract $\mathcal{C} = (\mathcal{J}, \mathcal{F}, \mathcal{U}, \mathcal{T}, \mathcal{V})$ specifies requested joint names, required fields, native joint units, one declared clock with continuity/span rules, and applicable producer-validity requirements. The adapter maps source columns or arrays into these declarations; the shared validator does not infer axes from array order. Numerical finiteness authorizes a calculation, not a claim of physical measurement validity. A producer flag can withhold that authorization, whereas an absent flag is reported as unspecified rather than silently asserted true.

The executable procedure is:

1. Resolve unique requested names and check compatible arrays. Check only the fields required by the property; position-only eligibility does not require velocity.
2. Retain missing/nonfinite values and applicable producer vetoes as reason-coded unavailability. Do not zero-fill or drop a row and reconnect the sequence across it.
3. Check strictly positive intervals in the declared clock, maximum gap and minimum observed span. Break residence at unavailable inputs or discontinuity. Compute integral residuals only when required fields and native units are available.
4. Return each supported numerical result, a true/false predicate result where evaluable, or an undetermined result with its missing prerequisites. Keep acquisition freshness and physical motion undetermined when their independent evidence is absent.

The standard-library implementation and four adapters are included in Online Resource 2. Per-joint residuals retain native angular or linear units; mixed units are not combined into an angular norm. The supplied implementation computes record properties and withholds unsupported physical claims. It is not a general device-acquisition verifier or a controller deployed on the robot.

We compare recorded-age cutoffs, the archived configured speed bound, position-only small span, whole-message finite q, v, required-arm finite q, v, and producer-aware required-field checks. Each comparison uses the same retained rows or windows for its stated property. The age predicate uses the saved HAL-to-receiver age, rather than claiming to reproduce an entire ROS timestamp diagnostic. Message-wide finiteness is an explicit conservative baseline, not a policy prescribed by ROS. Producer-aware checks use flags only where the source schema defines their semantics.

To control observation horizon, every incident and contextual sequence is partitioned into nonoverlapping 0.5 s reception-time bins anchored at its first sample. A bin requires at least 0.4 s observed span, finite required fields and positive gaps no larger than 0.1 s. Blocks, topics, arms and archives are never bridged. We retain all eligible bins and report the first held-case bin alongside the complete distributions; no interpolation or threshold fitting is used. Gap limits of 0.05/0.5 s, age cutoffs of 10/20/50/100 ms, and position tolerances of $0, 10^{-6}, 10^{-5}, 10^{-4}$ rad provide sensitivity checks. These comparisons quantify coverage and record-level disagreements, not errors against unavailable fault labels.

Software portability is tested by reusing the same validator through all four adapters. Twenty-seven checks on real records and explicitly transformed copies test joint-order invariance, scope invariance to unrelated nonfinite fields, required-field deletion, validity vetoes, missing-unit withholding and time-gap handling. The transformed copies are implementation fixtures, not injected physical faults or additional empirical trials. Hardware transfer and calibrated performance require different evidence.

3.6 Evidence views and descriptive comparators

For the incident, recency, payload residence, internal consistency and execution events provide complementary evidence views. The equations below define the numerical properties used by the shared contract.

Let $b(i)$ be the first index in the current uninterrupted sequence of identical kinematic vectors. Its observed residence is

$$H_i = (r_i - r_{b(i)})10^{-9}. \tag{4}$$

A candidate inconsistency flag is present when $H_i \geq \tau$ and $\max_j |v_{ij}| > 0.25^{\circ}\mathrm{s}^{-1}$. The gate is inherited from the configured stationary threshold. This exploratory flag localizes a property of a saved trace. It was not installed on the robot, and its first occurrence measures time from the first observed repeated value, not latency from an independently known fault onset.

For joint j, the integral residual over a window is

$$E_j = \sum_{i=1}^{n-1} \frac{v_{i+1,j} + v_{i,j}}{2}(t_{i+1} - t_i) - (q_{n,j} - q_{1,j}), \tag{5}$$

using trapezoidal integration and receiver-relative seconds t_i. This is an internal consistency measure. Both terms originate from the same feedback chain; neither is an external motion reference. Noise, filtering, quantization, timing offsets and different field update histories may all affect E_j. We report its value without fitting a physical fault model or assigning a universal alarm threshold.

To retain a holding comparison, we extract per-arm position ranges and velocity extrema from each step's stored channel summary. For a position-only rule,

$$B_q(\epsilon) = \mathbb{1}\left[\max_j \{\max_i q_{ij} - \min_i q_{ij}\} \leq \epsilon\right]. \tag{6}$$

These summary extrema can establish small observed range but cannot reconstruct ordering or repeated-frame duration. We keep their evidence level distinct from the six raw CSV windows. Completed steps are contextual comparisons, not certified healthy controls.

3.7 Sensitivity analysis and numerical verification

The initial incident plan followed source inspection and was not prospectively preregistered. Its complete age, dwell, tolerance, lag and decimation grids remain in Online Resource 1 and the parameter files. Every first, second or fifth sample was retained without filling gaps. Tolerances were applied separately in rad and rad/s, without fitting an operating point.

We recomputed the integral residual with receiver, HAL and header time differences. Elbow position differences over lags of 1, 2, 5 and 10 samples were compared with midpoint reported velocity. Each lag uses actual elapsed time, rather than assuming a perfect 100 Hz stream. No smoothing was applied. The independent numerical check used the standard-library CSV reader and decimal arithmetic to verify the 67-vector identity and constant-velocity integral. Raw row counts and velocity peaks were also compared against the independently saved per-step summaries.

Representative original MCAP payloads were decoded with two implementations of Common Data Representation (CDR), and their named fields and header stamps were compared. An independent standard-library check recomputed field masks and exact residence from the raw extraction, and parsed the original bench CSVs. These checks verify extraction and arithmetic, not physical sensing.

All analyses ran locally in Python 3.12.3 with NumPy 1.26.4 and pandas 2.3.3. Figures were generated programmatically with Matplotlib. The scripts, parameter file, source-data tables and complete sensitivity outputs are supplied as Online Resource 2. Online Resource 1 describes

the data dictionary, provenance and interpretation boundaries.

3.8 Statistical scope

The incident setting is one robot-day; S1/S2 contribute a second robot-day with a different task mode and SDK. The MCAP audit adds archive records with storage identities, not certified physical replications. Sessions, archives, byte blocks, sampled intervals, topic views and frames are nested and temporally dependent. These convenience-selected sources are not exchangeable experimental groups. We report exact archive counts, ranges, empirical quantiles and within-window contrasts. No population confidence intervals or hypothesis tests are reported because the selection mechanism and independent replication do not support them. Neither the 1,968 incident rows nor the additional arm-topic views are independent experiments. The archive supplies no verified fault-onset labels, independent physical motion measurement or balanced normal/fault test set. We consequently do not estimate sensitivity, specificity, false-positive rate, root-cause accuracy or policy task-success rate.

4 Results

4.1 The archive contains distinct exception paths

R1 and R4 each failed at their first step with a bilateral-goal acceptance error. Driver journals record that the right trajectory controller was not running, while the application attempted to cancel the already accepted counterpart. R2 step 5 and R3 step 4 instead failed after their earlier steps had completed, with a recorded velocity-limit exception. R6 step 3 was rejected before command execution with the message that the robot was moving. Its preceding step had a successful trajectory-result event. These paths separate controller readiness, observed velocity excursion and pre-step feedback interpretation; they cannot be collapsed into one physical failure class.

Figure 1 shows the right-elbow records for the three steps with continuous telemetry. Table 2 reports all bilateral windows. No position or velocity entries were missing in those CSVs, and all six windows had strictly increasing source, header and reception timestamps. Their availability does not fill the missing raw traces for the other 27 steps.

4.2 Advancing timestamps coexist with held kinematic feedback

In R6 step 3, all 67 right-arm kinematic vectors were identical at stored precision across 0.663923 s. There were 66 adjacent equal pairs, 67 distinct HAL timestamps and 67 distinct header timestamps. The median right-arm interarrival interval was 10.078 ms, and the maximum was 19.103 ms. Recorded right-arm age had a median of 0.315 ms and a maximum of 17.469 ms. Every row passed 50, 100 and 500 ms recorded-age cutoffs. Timestamp progression and bounded age thus coexisted with unchanged kinematic content in this window.

The left arm supplied a contemporaneous contrast: 68 distinct kinematic vectors in 68 rows over 0.672711 s, with maximum age 16.683 ms. Its position range was small, but its full position–velocity payload changed. Figure 2 separates recorded age from payload residence, making the right-arm discrepancy visible without treating either arm as an independently labeled physical control.

Update behavior also differed across field groups. All seven right-arm external-effort fields each took two values; position, velocity, effort, current and temperature fields each remained constant. The 42-component signal vector therefore had two distinct values, whereas its 14-component kinematic subvector had one. The longest unchanged interval for all 42 signals was 0.593470 s, compared with 0.663923 s for kinematics. This rules out describing the saved signal payload as entirely unchanged. It does not identify where the field histories diverged or establish that either external-effort value was newly acquired from a device.

4.3 Position–velocity disagreement strengthens the symptom description

The held right-elbow position was accompanied by a constant reported velocity of $1.897607^\circ\mathrm{s}^{-1}$. Over the observed receiver-time interval, this velocity implied 1.259864° of displacement, while recorded position displacement was zero. All other right-arm joints also had fixed position and velocity values; the elbow produced the largest integral

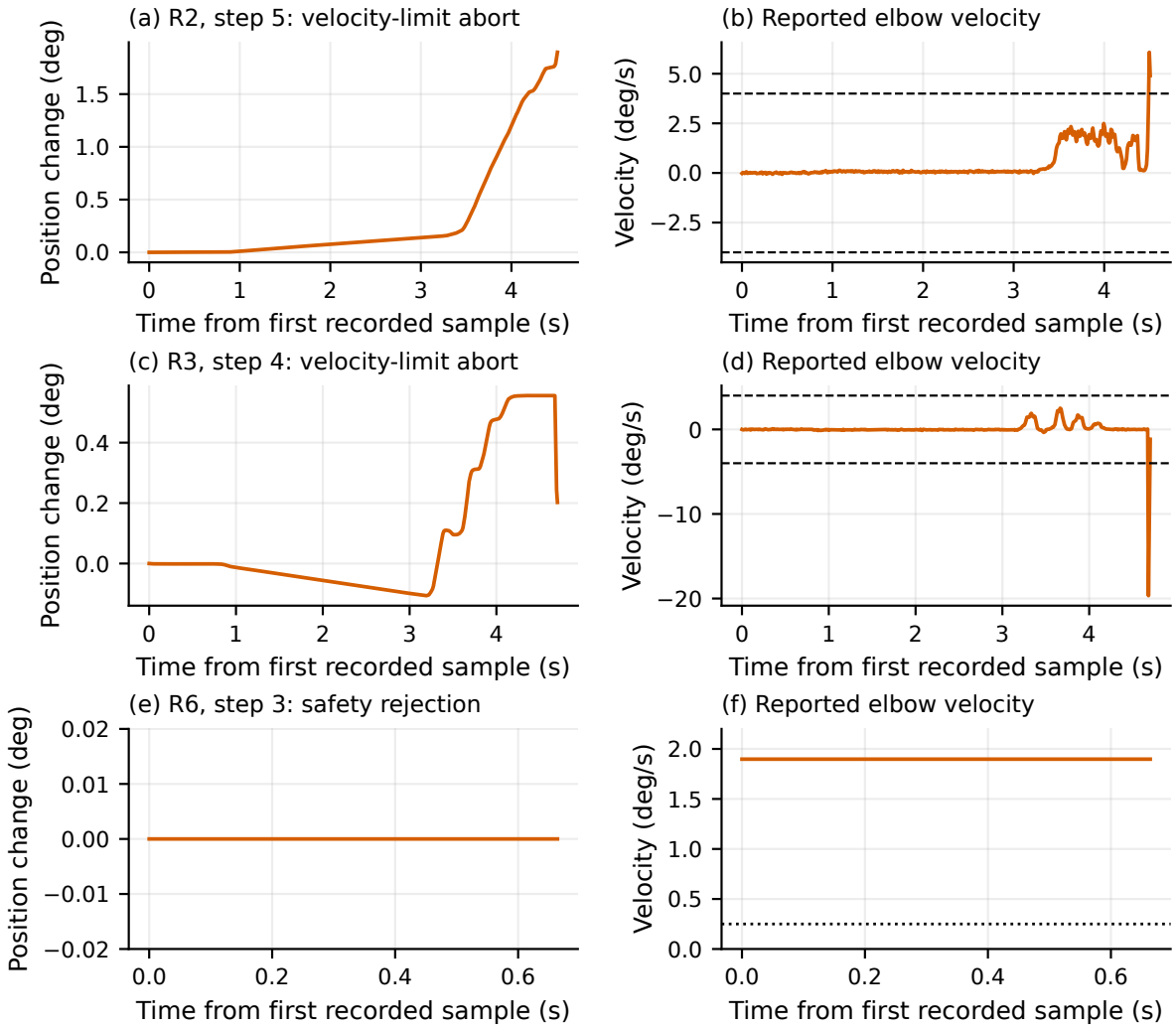


Fig. 1 Recorded right-elbow trajectories in the three selected steps. Panels (a), (c) and (e) show position relative to each window's first value; (b), (d) and (f) show reported velocity. Dashed horizontal lines in (b) and (d) indicate the configured $\pm 4°s^{-1}$ hard limit; the dotted line in (f) indicates the $0.25°s^{-1}$ stationary threshold.

residual. The left-elbow residual over its own window was $-0.002133°$. These quantities describe the record's consistency, not an externally measured displacement error.

The two earlier abort windows behaved differently. R2 and R3 right-elbow velocity peaks were 6.081914 and $19.664209°s^{-1}$, respectively, and their position values changed. At a one-sample difference lag, the corresponding maximum absolute position-derived elbow velocities were 5.848 and $16.109°s^{-1}$. A position change therefore accompanied each velocity excursion; the evidence is stronger than a spike confined to the velocity field. However, position and velocity came from the same feedback chain, so agreement does not independently validate physical speed. Longer difference lags attenuated the peaks: at 10 samples the maxima were 2.033 and $3.514°s^{-1}$. This sensitivity argues against presenting a derivative peak without its differencing interval.

4.4 Small position ranges also occur in completed steps

R5 completed 16 steps before its session status became stopped. In every one of those steps, the

Table 2 Recomputed telemetry measures. Age is reception time minus the recorded HAL timestamp. Unique values refer to the complete 14-component position–velocity vector. E_{elbow} is the signed integral residual in Eq. 5; duration and residual use receiver time

Step	Arm	Rows	Duration (s)	Unique vectors	Maximum age (ms)	Peak $\lvert v\rvert$ ($^\circ s^{-1}$)	E_{elbow} ($^\circ$)
R2, step 5	Left	449	4.512	449	12.003	1.548	0.0053
	Right	448	4.508	448	12.169	6.082	−0.0015
R3, step 4	Left	468	4.703	468	10.785	2.094	−0.0020
	Right	468	4.703	468	13.360	19.664	0.0492
R6, step 3	Left	68	0.673	68	16.683	0.107	−0.0021
	Right	67	0.664	1	17.469	1.898	1.2599

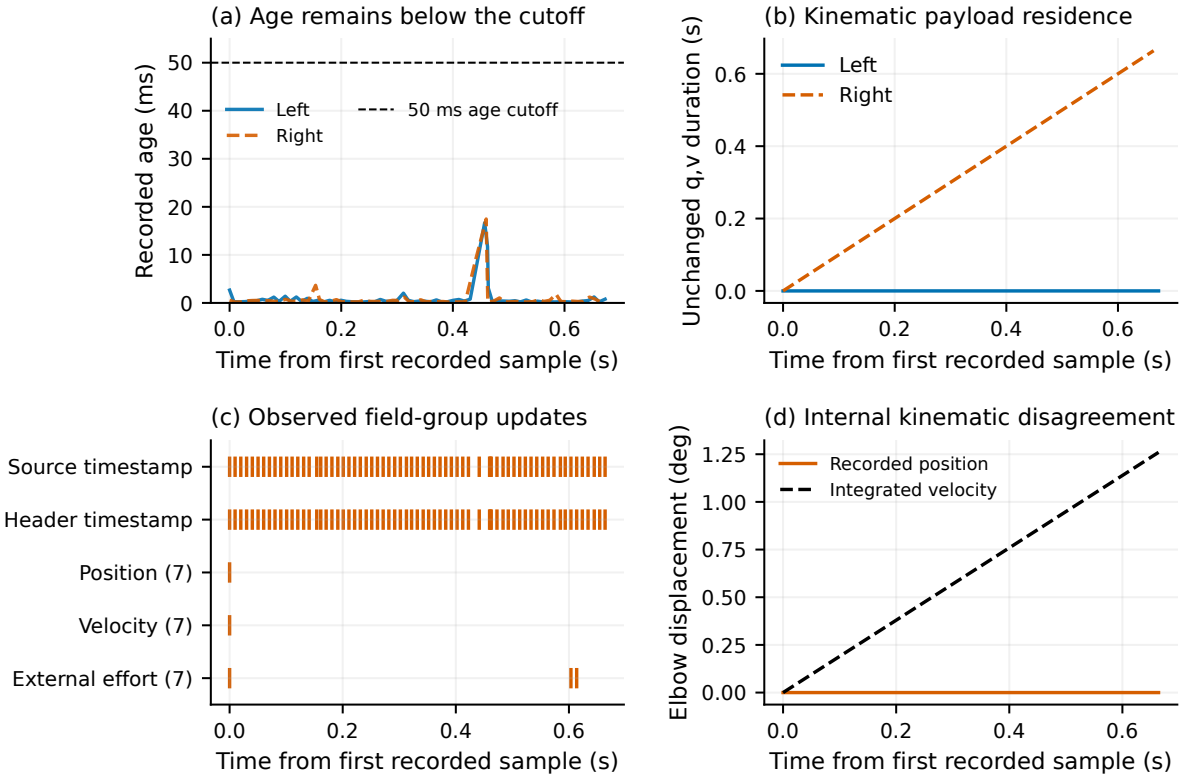


Fig. 2 Field-resolved evidence in R6 step 3. (a) Recorded age and a 50 ms descriptive cutoff. (b) Residence at an unchanged 14-component kinematic vector. (c) First values and observed numerical updates in the right-arm field groups; ticks represent changes, not verified acquisition events. (d) Elbow displacement obtained from position and from integrated velocity. The two arm windows use their own first received sample as time zero, so the display is a within-window comparison rather than sample-level bilateral synchronization

left-arm summary reported a maximum joint position range between 1.198×10^{-6} and 1.465×10^{-6} rad. A position-only rule with $\epsilon = 10^{-5}$ rad would mark all 16 left-arm steps as small-span windows. The corresponding maximum reported speeds ranged from 0.107 to 0.164$^\circ s^{-1}$. The right arm moved over substantially larger recorded ranges (Fig. 3). This supports a holding interpretation for the left-arm summaries and demonstrates that small position range alone is an ambiguous diagnostic feature. Without raw left-arm traces or independent physical labels for R5, these observations cannot be converted into a measured false-positive rate.

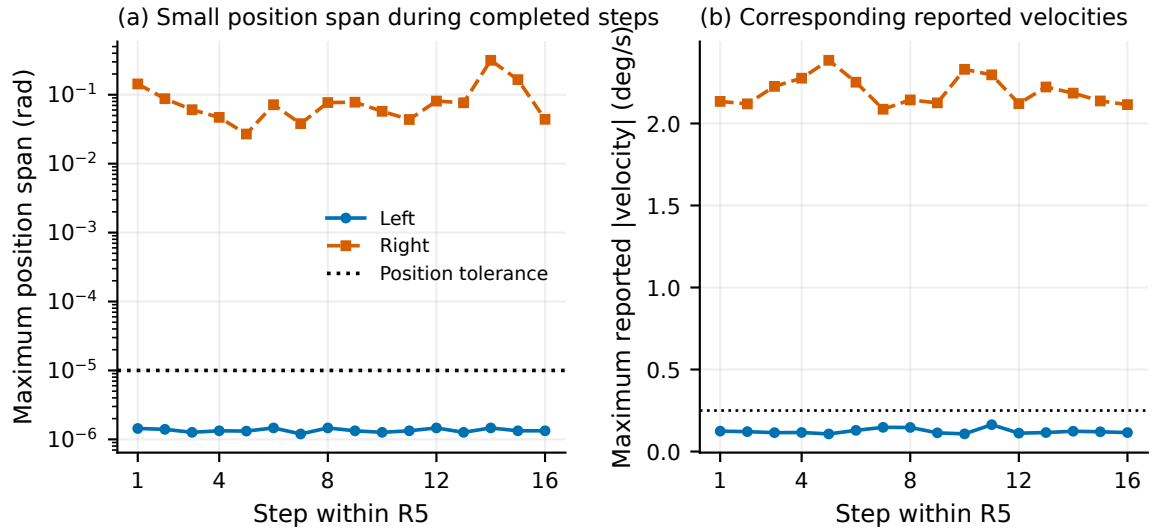


Fig. 3 Stored per-step extrema for the 16 completed R5 steps. (a) Largest observed joint position range per arm; the dotted line is the 10^{-5} rad position-only tolerance. (b) Maximum absolute reported joint velocity; the dotted line is the configured stationary threshold. These are archived summaries and do not reconstruct missing continuous trajectories or establish task success

4.5 Sensitivity checks preserve the evidence distinction

The R6 right-arm vector remained exactly equal, so its equality finding was unaffected by tolerances up to 10^{-4} in the stated units. Positive age cutoffs below the largest recorded delay rejected some rows; cutoffs of 50 ms or more accepted all rows in both arms. Tightening a recency cutoff can reject delays without establishing whether a passing kinematic value is newly measured (Fig. 4a).

At a 0.25 s residence threshold, the first flag occurred at 0.2513 s with all rows retained. Every-second and every-fifth subsampling gave 0.2615 and 0.2513 s (Fig. 4b). The complete dwell grid is retained in Online Resource 2. None of the other five raw arm windows contained any exactly repeated adjacent kinematic vectors. These observations establish robustness on the saved traces, not performance over unseen normal or faulty operation. Decimation changes observation opportunities and is not a model of transport packet loss.

Changing the integration clock left the held-window conclusion intact: the right-elbow residual was 1.2599° using receiver time and 1.2604° using HAL or header time. By contrast, short-lag finite differences altered the numerical peaks of the two excursion windows. Both results are retained in Online Resource 2 rather than selecting one favorable time-base or lag setting.

4.6 Another operational context exposes missingness and brief repetition

The 32 blocks contain 90,451 complete JSON records with no parse errors. Direct group-state topics supply 17,617 arm samples; the aggregated topic supplies 17,708 arm views of the same system (Table 3). Every extracted position vector is finite. In S2, 2,067 direct samples and 2,068 aggregated arm views have empty velocity arrays. Those direct messages still have advancing recorded clocks and HAL ages below 15.204 ms. Message recency therefore does not establish that all fields required by a stationary-state or consistency check are available.

No fully observed direct kinematic vector repeats in adjacent samples. The aggregated topic contains 95 adjacent equal pairs, with maximum observed residence 12.770 ms. Neither view produces a nonzero-velocity residence flag at 0.10, 0.25 or 0.50 s. Brief repetition in this additional context shows why equality and sustained inconsistent residence are different observations; the result is not an estimate of healthy specificity. An exploratory within-block comparison matched

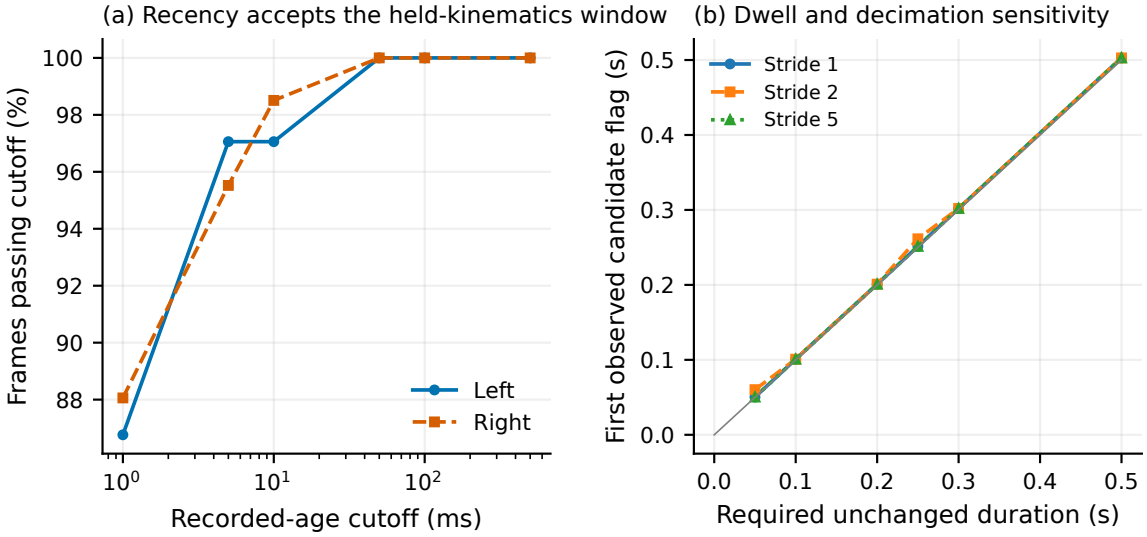


Fig. 4 Deterministic sensitivity checks for R6 step 3. (a) Fraction of saved rows satisfying each recorded-age cutoff. (b) First observed candidate inconsistency flag versus required kinematic residence for three deterministic decimation strides. The diagonal is an ideal observation-time reference; flags are not measured fault-detection latencies. No error bars are shown because these are exact functions of selected dependent traces

identical direct vectors within ±50 ms of reception for 15,605 of 15,640 finite aggregated kinematic rows, further documenting the numerical dependence of these views.

Of 349 eligible direct position windows, 311 have complete velocity and 38 do not. In the latter, the maximum within-window joint position span ranges from 0.000824 to 17.652°. Zero-filling their absent velocity would manufacture a standstill estimate while recorded positions vary. Among the 311 fully observed windows, 304 satisfy the 10^{-5} rad small-span rule. Their maximum absolute joint integral residual is 0.00934°. Applying the same first-0.5-s rule to R6 yields 50 samples spanning 0.492742 s and a 0.93503° residual. This duration comparison characterizes the recorded discrepancy; different modes, versions and unobserved acquisition processes prevent using the contrast to calibrate detector accuracy.

The 0.05, 0.10 and 0.50 s gap limits retain the same 311 direct and 302 aggregated finite-velocity windows. Figure 5 shows the missing-field regime, an illustrative position trace and the finite-window residual distributions. The 38 direct windows without velocity remain undetermined for this calculation, rather than being silently removed from the evidence inventory.

4.7 Validation scope changes numerical eligibility

All 28 selected archives were read with stable observed metadata; 27 yielded joint-state messages in the sampled intervals. The 77,627 decoded messages provide 155,254 dependent arm views. Table 4 retains every selected record, including empty samples. These sources broaden the schema audit without establishing independent physical replications.

Both selected arms have finite position and velocity in all 77,627 messages. However, 54,897 contain nonfinite velocity elsewhere. A message-wide check therefore retains 22,730 messages, compared with 77,627 for the two named arms (Fig. 6a). This difference concerns numerical eligibility for an arm-level calculation. Other robot properties may require the unavailable joints. Together with S1/S2, the comparison distinguishes missing required velocity from missing values outside the requested joint set.

Exact arm-vector residence reaches 116.930 ms (Fig. 6b). Six runs exceed 100 ms while message-header time progresses. Such repetitions describe the record and do not provide physical fault labels. Complete equal-pair counts, clock relationships and gap/residence settings are retained in Online Resource 2.

Table 3 Additional context, kept separate from the incident archive. Rows are seven-joint arm views, not independent physical measurements. Equal pairs require finite position and velocity and a positive interarrival gap no greater than 0.1 s. Position and finite-velocity windows use the eligibility rules in Sect. 3.2

Archive	Topic view	Arm rows	Missing velocity	Equal pairs	Max hold (ms)	Position windows	With velocity
S1	Direct group	7,370	0	0	0	144	144
S1	Aggregated	7,412	0	46	12.770	140	140
S2	Direct group	10,247	2,067	0	0	205	167
S2	Aggregated	10,296	2,068	49	12.709	200	162

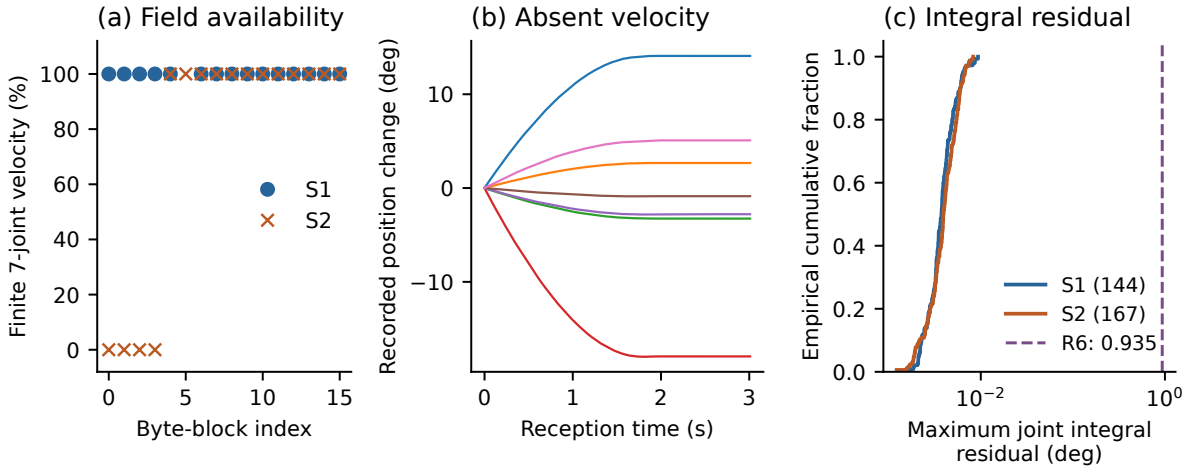


Fig. 5 Context extension from preselected, discontinuous byte blocks. (a) Fraction of direct arm samples with finite seven-joint velocity; markers are not connected across blocks. S1 block 5 contains no eligible arm message. (b) All seven left-arm position changes in S2 block 1, the first nonzero-offset S2 block; every velocity array is empty. Colors distinguish joints. These are reported positions, not independent motion measurements. (c) Empirical distributions of maximum absolute joint integral residual in eligible direct windows (0.4–0.5 s observed span). The dashed line is the duration-matched R6 interval.

The bench audit adds 58,493 rows from 3 distinct archived records. 6 finite, zero-velocity rows carry producer flags declaring velocity unavailable. Each record also has a duplicate cycle–time pair with changed velocity/validity contents. Producer flags can therefore withhold velocity evidence even when $C_a = 1$ (Online Resource 1).

4.8 Paired predicates expose different evidence gaps

Table 5 gives the paired comparisons. A 20 ms recorded-age rule accepts all 1,968 incident rows, including the held inconsistent right-arm trace. The configured speed bound is exceeded in two of the six detailed arm traces; the held trace's peak of $1.898°s^{-1}$ remains below its $4°s^{-1}$ bound. These predicates evaluate timing and excessive reported speed, respectively. Neither output establishes that the kinematic fields are mutually consistent.

At the common horizon, 38 incident bins and 311 direct contextual bins are eligible for integral comparison (Fig. 7). The first held-case bin spans 0.492742 s and has residual 0.935 degrees; the maximum among the direct contextual bins is 0.00934 degrees. The 38 additional position-eligible context bins lack required velocity and receive no residual. Their missingness cannot be repaired by observing recent positions. Of the 311 finite-velocity contextual bins, 304 satisfy the 10^{-5} rad position-only small-span rule. This is a recorded holding descriptor, not evidence of a corresponding false-positive rate. The complete window set and threshold grid remain in Online Resource 2.

The validity comparisons show opposite effects. Whole-message finiteness withholds 54,897

Table 4 Archived JointState audit by storage group. Selected/read counts include every metadata-selected record; read means unchanged observed size/mtime around extraction. Message counts concern the selected intervals or complete small archive. The two eligibility columns count messages: all named joints versus both seven-joint arms. Storage groups, records and arm views are not independent physical trials. A zero residence indicates no eligible adjacent equality, and does not certify acquisition

Storage	Records selected/read	Messages	All joints finite q, v	Both arms finite q, v	Max arm residence (ms)
G1	2/2	1,500	0	1,500	0.000
G2	1/1	10,836	0	10,836	116.930
G3	21/21	42,561	0	42,561	0.000
G4	2/2	6,529	6,529	6,529	0.000
G5	2/2	16,201	16,201	16,201	0.000

of 77,627 messages (70.72%) despite finite inputs for both requested arms. Producer-aware checks instead withhold six of 58,493 numerically finite bench rows because their velocity flags deny availability. Thus, the contract neither maximizes acceptance nor equates acceptance with trust: it retains exactly the inputs required by the declared property, subject to the source's applicable vetoes. All 27 software invariant checks pass across the four adapters. This demonstrates portability of validation mechanics across the retained schemas; it does not demonstrate cross-platform fault discrimination.

4.9 An evidence contract limits diagnostic escalation

Table 6 maps each diagnostic statement to its minimum evidence. The recency view establishes progressing recorded clocks. The payload view adds exact kinematic repetition. The consistency view adds disagreement between recorded position and velocity, and distinguishes this symptom from a small holding range. Controller events establish the phase and software response. None of these views supplies the missing device acquisition provenance or independent physical motion reference.

The R6 control-event log places the preceding successful trajectory result approximately 0.094 s before the first right-arm row in the selected rejection window. It therefore establishes that a success event preceded the contradictory feedback window. It does not show that the robot physically stopped between those events, or that the next-step rejection was physically unnecessary. Similarly, recorded cancellation handling after the R2 and R3 exceptions establishes a software response; it does not establish the physical stop time.

5 Discussion

5.1 Diagnostic observability depends on signal semantics

The records expose three limits of message-level validation. Present fields can remain internally inconsistent despite recent timestamps; a recent message can omit a required velocity or mark it unavailable; and a nonfinite value outside the required joints can cause an unnecessarily broad rejection. The named-arm comparison makes the third distinction explicit. These observations support associating validity with the property, joint set and publication layer under examination.

This result qualifies, rather than challenges, the value of combining observability data [2]. Cross-layer records explain which operation failed and when a protection path was entered. They do not necessarily add independent measurements of the state that triggered that path. Tracing and runtime verification can expose that provenance if it is instrumented [4, 5]; the current record illustrates what remains ambiguous when it is absent. The empirical addition is the combination of advancing HAL and message timestamps, held 14-component kinematics, partially changing auxiliary fields and an application-level rejection within one traceable operational record.

Here, *diagnostic observability* means the ability to distinguish explanations from a specified record. It is not a claim about Kalman observability of a dynamical model or a rank condition for the robot. Likewise, data trustworthiness is claim-specific: the same timestamp can be adequate for

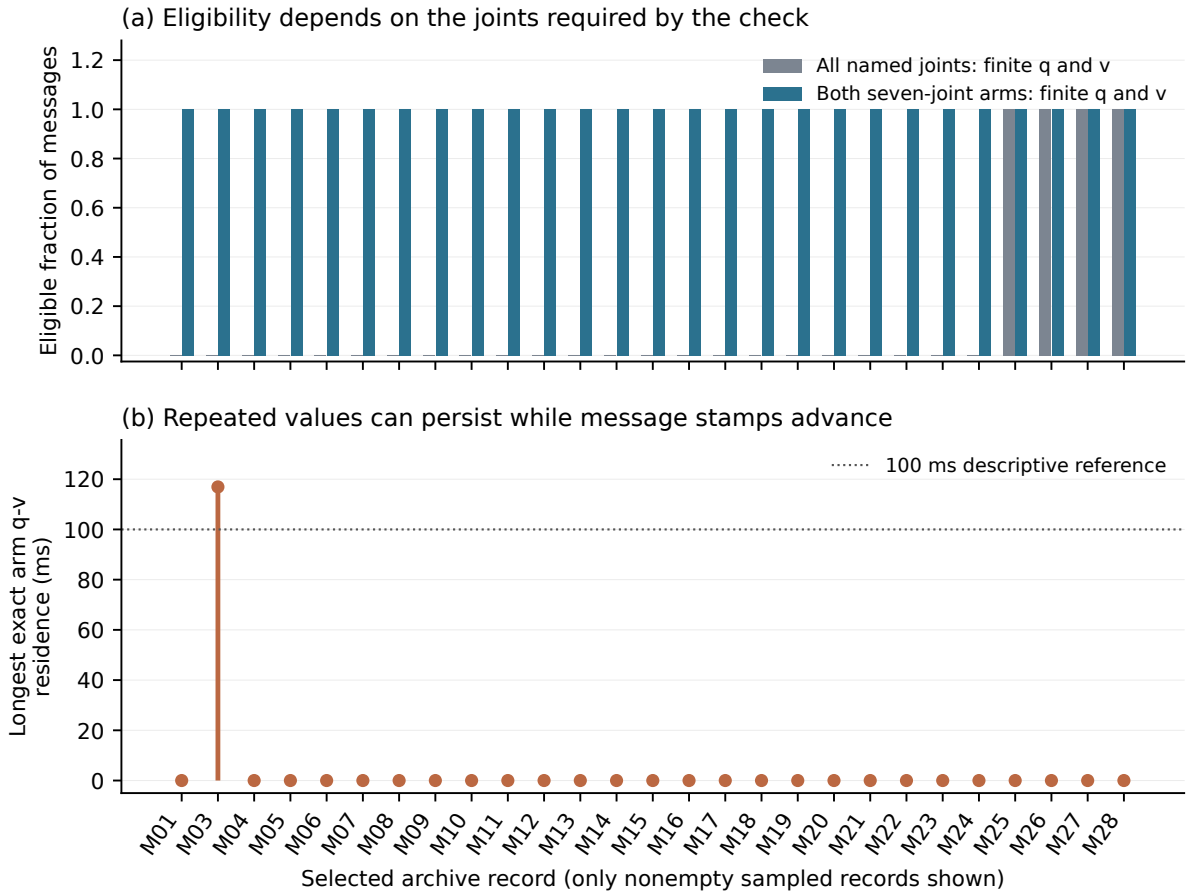


Fig. 6 The fields required by a diagnostic determine numerical eligibility. (a) Message-wide versus both-arm finite position–velocity checks in every selected record with decoded messages. The denominator is the same message set for both checks. (b) Longest exact seven-joint arm-vector residence per record, breaking at missing fields, nonpositive intervals and gaps above 0.1 s. The dotted 100 ms line is a descriptive reference. Header progression and numerical repetition describe the stored record; they do not supply physical fault labels. Unavailable records and empty sampled archives remain in Table 4 and the complete selection ledger

ordering application events and inadequate for proving the age of a physical measurement.

5.2 Increment over individual monitoring predicates

The comparison identifies a practical role for the contract before an existing monitor evaluates its property. Timestamp diagnostics can remain useful for delivery timing while the contract withholds acquisition-freshness claims. A speed bound can enforce its stated limit while a separate internal-consistency calculation exposes a different symptom. A runtime-verification engine can consume these property-specific outputs with explicit unavailability rather than a substituted zero. This composition does not replace the existing tools or introduce a new temporal semantics.

The paired comparison identifies two opposing validity decisions. A whole-message check can discard finite required-arm fields; a finite-only check can admit a producer-declared unavailable value. One global verdict would conceal that distinction. The four-adapter implementation makes these premises inspectable across representations, while physical fault recognition remains untested.

5.3 Why root cause remains unidentifiable

An elementary indistinguishability argument makes the boundary precise. Let the observable

Table 5 Paired monitoring-predicate comparisons on unchanged records. Each row states its own denominator and property. Counts describe outputs or eligibility, not correctly classified physical states. Different rows are not summed into a single performance score

Predicate	Observed output	Contract-based interpretation
Recorded age ≤ 20 ms	1,968/1,968 incident rows pass	Kinematic repetition and inconsistency remain separately evaluable
Configured speed bound	2/6 incident arm traces exceed $4^{\circ}\mathrm{s}^{-1}$	Held R6 right remains below the bound; excessive speed and inconsistency differ
Position span $\leq 10^{-5}$ rad	304/311 finite-velocity direct context bins meet the rule	Small range alone does not supply a physical fault label
Required q, v for residual	311/349 direct context bins eligible	38 remain undetermined; no velocity imputation
Whole-message vs named arms	22,730 vs 77,627 eligible messages	54,897 additional arm-eligible messages; no claim for the unavailable joints
Finite vs producer-aware q, v	58,493 vs 58,487 bench rows eligible	Six unavailable zero-velocity rows withheld despite finite numbers

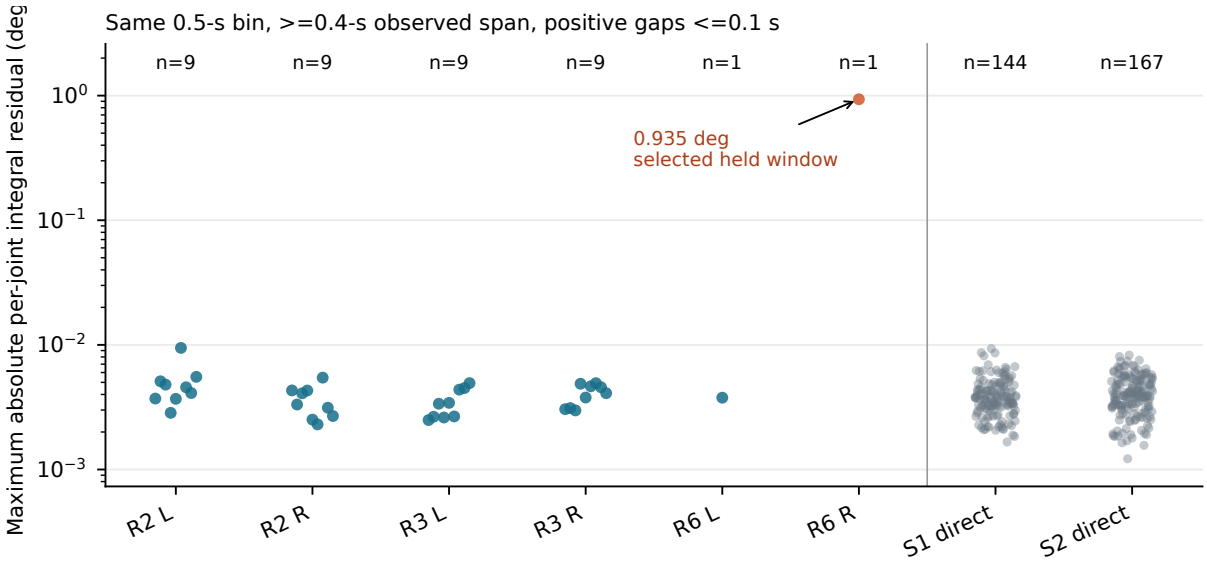


Fig. 7 Integral discrepancy at a common observation horizon. Every point is an eligible 0.5 s bin with at least 0.4 s observed span and positive reception gaps no larger than 0.1 s. The ordinate is the largest absolute joint residual in each bin, displayed in degrees. All 38 incident bins and 311 direct contextual bins are retained; the held right-arm trace contributes one eligible bin. Contextual aggregate-topic views are excluded here because they depend on the direct view. Counts denote nested windows, not independent trials. Separation in this retained sample is descriptive and is not a calibrated physical-fault threshold

record be $\mathcal{O} = \{s_i, h_i, r_i, \mathbf{q}_i, \mathbf{v}_i, \mathbf{z}_i\}$, where $\mathbf{z}_i$ denotes ancillary fields and events. Consider two possible mechanisms. In mechanism A, each cycle acquires a new device or estimator output, but quantization, bias or a sensor-side defect yields the same kinematic values; ancillary fields update as recorded. In mechanism B, a downstream layer reuses an earlier kinematic value and assigns current timestamps, while ancillary fields follow the same recorded sequence. The unobserved acquisition identity differs, but both mechanisms can produce the same $\mathcal{O}$.

Any decision based solely on $\mathcal{O}$ must give the same answer in both mechanisms. It therefore cannot, without additional assumptions, uniquely identify downstream caching or prove per-field acquisition freshness. The nonzero-velocity integral residual rules out interpreting the saved position and velocity as a mutually consistent exact

Table 6 Evidence contract for diagnostic statements. Undetermined denotes missing discriminating evidence, not an inferred normal condition. Evidence requirements are claim-specific rather than an ordinal quality score

Statement	Evidence required	Verdict in this archive
Recorded timestamps progress	Ordered timestamps and positive differences	Established for the six raw arm windows
Required joint values are numerically present	Requested names, compatible array lengths and finite position/velocity values	Depends on the joint set; see Table 4. Producer flags and acquisition validity require separate checks
Kinematic payload is held	Finite fields, distinct vectors and residence	Sustained in R6; shorter repeats in contextual aggregate and MCAP records
Recorded kinematics disagree	Position increments and integrated velocity	Established internally in the held window
Execution was rejected or aborted	Phase-specific application and controller records	Established for the named software paths
A new device measurement was used	Original acquisition identity/time and field validity	Undetermined; HAL/message timestamps are insufficient
The robot moved or stopped physically	Synchronized independent motion observation	Undetermined
A particular cache or hardware fault caused it	Discriminating cross-layer command/feedback evidence	Undetermined

kinematic history. It does not select which mechanism generated the inconsistency. This is an analytical counterexample, not a simulation dataset or an estimate of how often either mechanism occurs.

5.4 Implications for recording and monitoring

The analysis supports a concrete recording contract. A device acquisition identifier should remain associated with the field group it qualifies, accompanied by acquisition time, validity status and publication time. If a layer emits a cached field, it should preserve or explicitly mark the earlier acquisition identity rather than imply a new sample through a new publication timestamp. When the acquisition identity is unavailable, downstream consumers should retain an undetermined freshness state instead of deriving a physical conclusion from recency alone.

Controller success, stationary-state admission and permission to start the next observation/action cycle require separate evidence and thresholds. Evaluating a settling or admission policy would additionally require valid feedback and an independent physical stop reference.

5.5 Limitations and discriminating next evidence

The persistent inconsistency remains one selected case. Equal-duration windows improve comparison but do not add independent incidents. Four-adapter checks establish schema portability, not hardware transfer or fault discrimination. The simple predicates are interpretable comparators rather than optimized detector baselines. The second robot-day has different task, SDK and capture conditions, preventing causal cross-robot or version inference. Only six of 11 historical incident directories have copied artifacts. The context sample is fixed in byte space, potentially overweighting high-volume phases. MCAP intervals use another convenience design and cannot certify every archive's physical origin. These sources support recorded-property comparisons, not prevalence or diagnostic performance.

R5 has summaries without copied continuous traces. Context records do not certify physical health. Source snapshots and configuration digests establish traceability rather than exact historic binary behavior. Physical scene, contact history, final hardware commands and independent stop measurements are incomplete. These gaps prevent causal attribution and safety acceptance while leaving the numerical record audit reproducible.

The most informative next evidence is synchronized acquisition identity, final device command and an independent motion reference during both ordinary holding and the same exception paths. This would distinguish legitimate repeated measurements, sensor-side faults and downstream republishing. Repeated, labeled sessions across platforms would then be needed to estimate diagnostic performance with session-level uncertainty. Additional unlabeled rows alone cannot resolve the acquisition or physical-state ambiguity.

6 Conclusion

A reusable field-specific contract separates numerical eligibility, producer availability, clock continuity and internal kinematic evidence before drawing a diagnostic conclusion. Paired retrospective comparisons show why these premises matter: recent and below-bound feedback can remain inconsistent; a message-wide finite check can discard valid inputs for the requested arms; and finite zero velocity can still be declared unavailable. Equal-duration windows retain the selected inconsistency without extending observation time, while missing required velocity yields an explicit undetermined result. The shared implementation preserves these rules across four archived schema families. These results support reusable evidence validation ahead of existing monitoring predicates, without establishing physical acquisition freshness, a unique fault mechanism or cross-platform detection accuracy.

Supplementary information

Online Resource 1 (`ESM_1.pdf`) documents selection, provenance, schemas, sensitivity and implementation checks. Online Resource 2 (`ESM_2.zip`) supplies pseudonymized inputs, the shared validator and adapters, complete comparison outputs and figure-generation code. Original private logs, network identifiers and third-party full texts are excluded.

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Author contributions. Final authorship and contributions require author verification.

Data availability. Prepared supplementary resources contain pseudonymized inputs. Original records remain private. Authors will confirm final access conditions, license and deposit status before submission.

Code availability. Analysis code and parameters are included in prepared Online Resource 2.

Ethics and consent. This retrospective analysis used existing robot records without a new physical, human or animal experiment. Institutional requirements remain for author verification.